



Ecole Supérieure en Informatique

ESI-Sidi Bel Abbès

Lab Report 09

The implantation of an IDS/IPS

*Computer Systems and Network Security
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1. Objectives :

Gain hands-on experience in threat detection, rule creation, and IDS testing for real-world security monitoring scenarios.

2. Part 2 – IDS Mode

2.1 Procedure:

sudo systemctl stop suricata

sudo suricata -c /etc/suricata/suricata.yaml -i eth0

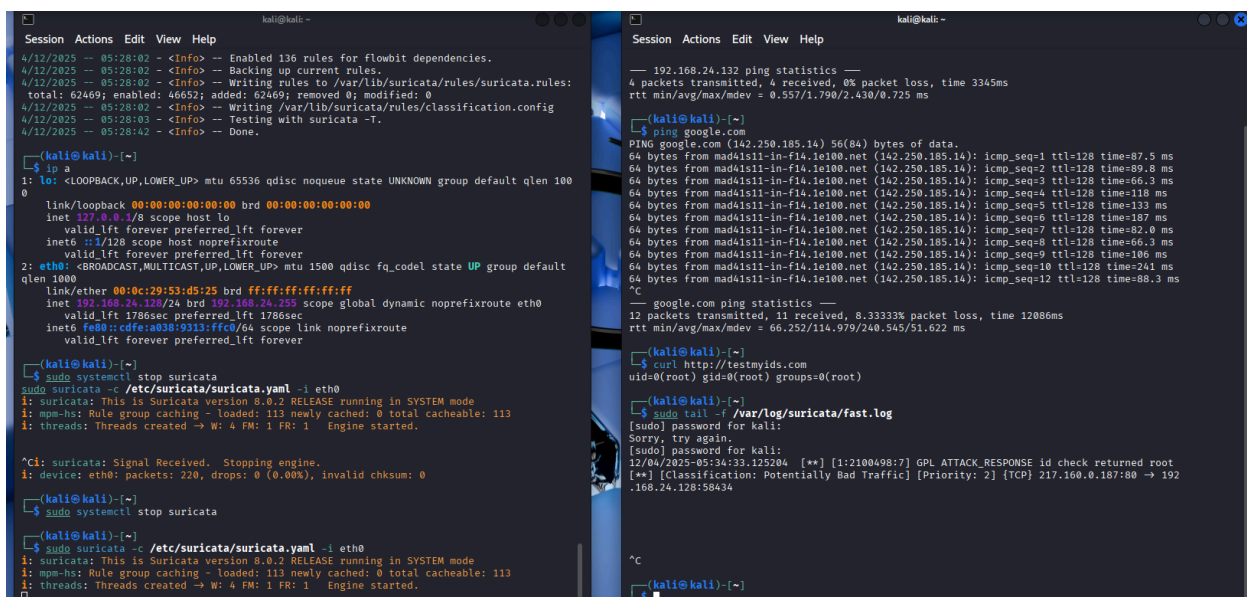
Generate traffic:

ping google.com

curl http://testmyids.com

2.2 Answers

2.3 Screenshoot :



The image shows two terminal windows from a Kali Linux machine. The left window displays the configuration of Suricata, including the IP address (192.168.24.128), the interface (eth0), and the ruleset path. It also shows the output of the 'systemctl stop suricata' command. The right window shows the output of the 'ping google.com' command, displaying the IP address (142.250.185.14) and the time taken for the ping (87.5 ms). It also shows the output of the 'curl http://testmyids.com' command, displaying the response from the server (200 OK) and the time taken for the curl (12086ms).

```
kali@kali:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:53:45:25 brd ff:ff:ff:ff:ff:ff
    inet 192.168.24.128/24 brd 192.168.24.255 scope global dynamic noprefixroute eth0
        valid_lft 1786sec preferred_lft 1786sec
    inet6 fe80::cde:a038:9313:ffc0/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

kali@kali:~$ sudo systemctl stop suricata
[sudo] suricata: version 8.0.2 RELEASE running in SYSTEM mode
1: nme-hs: Rule group caching - loaded: 113 newly cached: 0 total cacheable: 113
1: threads: Threads created -> W: 4 FM: 1 FR: 1 Engine started.

^C: suricata: Signal Received. Stopping engine.
1: device: eth0: packets: 220, drops: 0 (0.00%), invalid checksum: 0

kali@kali:~$ sudo systemctl stop suricata

kali@kali:~$ sudo suricata -c /etc/suricata/suricata.yaml -i eth0
[sudo] suricata: version 8.0.2 RELEASE running in SYSTEM mode
1: nme-hs: Rule group caching - loaded: 113 newly cached: 0 total cacheable: 113
1: threads: Threads created -> W: 4 FM: 1 FR: 1 Engine started.

kali@kali:~$ ping google.com
PING google.com (142.250.185.14) 56(84) bytes of data:
64 bytes from mad41s11-in-f14.1e100.net (142.250.185.14): icmp_seq=1 ttl=128 time=87.5 ms
64 bytes from mad41s11-in-f14.1e100.net (142.250.185.14): icmp_seq=2 ttl=128 time=89.8 ms
64 bytes from mad41s11-in-f14.1e100.net (142.250.185.14): icmp_seq=3 ttl=128 time=66.3 ms
64 bytes from mad41s11-in-f14.1e100.net (142.250.185.14): icmp_seq=4 ttl=128 time=118 ms
64 bytes from mad41s11-in-f14.1e100.net (142.250.185.14): icmp_seq=5 ttl=128 time=193 ms
64 bytes from mad41s11-in-f14.1e100.net (142.250.185.14): icmp_seq=6 ttl=128 time=187 ms
64 bytes from mad41s11-in-f14.1e100.net (142.250.185.14): icmp_seq=7 ttl=128 time=82.0 ms
64 bytes from mad41s11-in-f14.1e100.net (142.250.185.14): icmp_seq=8 ttl=128 time=66.3 ms
64 bytes from mad41s11-in-f14.1e100.net (142.250.185.14): icmp_seq=9 ttl=128 time=106 ms
64 bytes from mad41s11-in-f14.1e100.net (142.250.185.14): icmp_seq=10 ttl=128 time=241 ms
64 bytes from mad41s11-in-f14.1e100.net (142.250.185.14): icmp_seq=12 ttl=128 time=88.3 ms
^C
--- google.com ping statistics ---
12 packets transmitted, 11 received, 8.33333% packet loss, time 12086ms
rtt min/avg/max/mdev = 66.252/114.979/240.545/51.622 ms

kali@kali:~$ curl http://testmyids.com
uid=0(root) gid=0(root) groups=0(root)

kali@kali:~$ sudo tail -f /var/log/suricata/fast.log
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
12/04/2025:05:24:32.125204: [**] [1:2100498:7] GPL ATTACK_RESPONSE id check returned root
[**] [Classification: Potentially Bad Traffic] [Priority: 2] [TCP] 217.160.0.187:80 -> 192.168.24.128:58434

^C

kali@kali:~$
```

3. Part 3 — Rule Writing :

3.1 Procedure:

Step 6 — Create custom rule

File:

```
nano /etc/suricata/rules/custom.rules
```

Add this line:

```
alert http any any -> any any (msg:"Example.com Access Detected";  
content:"example.com"; sid:1000001; rev:1;)
```

Step 7 — Enable rule in suricata.yaml :

Open:

```
sudo nano /etc/suricata/suricata.yaml
```

Ensure this line exists:

```
- /etc/suricata/rules/custom.rules
```

Step 8 — Restart Suricata

```
sudo systemctl restart suricata
```

3.3 Screenshoot :

```

root@kali: ~
Session Actions Edit View Help
GNU nano 8.6 /etc/suricata/suricata.yaml *
#
# This parameter has no effect if auto-config is disabled.
#
hashmode: hash5tuplesorted

##
## Configure Suricata to load Suricata-Update managed rules.
##

default-rule-path: /var/lib/suricata/rules
rule-files:
- suricata.rules
- /etc/suricata/rules/custom.rules
##
## Auxiliary configuration files.
##

classification-file: /etc/suricata/classification.config
reference-config-file: /etc/suricata/reference.config
# threshold-file: /etc/suricata/threshold.config

##
## Suricata as a Firewall options (experimental)

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify

```

```

(kali@kali)-[~]
$ sudo tail -f /var/log/suricata/fast.log
12/04/2025-05:34:33.125204  [**] [1:2100498:7] GPL ATTACK_RESPONSE id check returned root [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 217.160.0.187:80 → 192.168.24.128:58434
12/04/2025-05:55:26.629748  [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 192.168.24.132:68 → 192.168.24.254:67
12/04/2025-05:55:26.629745  [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 192.168.24.132:68 → 192.168.24.254:67
12/04/2025-05:55:42.451977  [**] [1:2100498:7] GPL ATTACK_RESPONSE id check returned root [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 217.160.0.187:80 → 192.168.24.128:35964
12/04/2025-05:55:42.451980  [**] [1:2100498:7] GPL ATTACK_RESPONSE id check returned root [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 217.160.0.187:80 → 192.168.24.128:35964
12/04/2025-05:58:36.181329  [**] [1:1000001:1] Example.com Access Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.24.128:37478 → 23.220.75.232:80
12/04/2025-06:00:04.229317  [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 → 255.255.255.255:67
12/04/2025-06:00:04.229317  [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 → 255.255.255.255:67
12/04/2025-06:00:12.280214  [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 → 255.255.255.255:67

```

```
kali@kali: ~
Session Actions Edit View Help

(kali@kali)~$ sudo tail -f /var/log/suricata/fast.log
12/04/2025-06:01:01.064267 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 → 255.255.255.255:67
12/04/2025-06:08:38.098783 [**] [1:2000001:0] ICMP Echo Request Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.24.132:8 → 192.168.24.128:0
12/04/2025-06:08:39.020134 [**] [1:2000001:0] ICMP Echo Request Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.24.132:8 → 192.168.24.128:0
12/04/2025-06:08:39.942608 [**] [1:2000001:0] ICMP Echo Request Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.24.132:8 → 192.168.24.128:0
12/04/2025-06:08:40.863576 [**] [1:2000001:0] ICMP Echo Request Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.24.132:8 → 192.168.24.128:0
12/04/2025-06:08:41.785749 [**] [1:2000001:0] ICMP Echo Request Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.24.132:8 → 192.168.24.128:0
12/04/2025-06:08:42.707468 [**] [1:2000001:0] ICMP Echo Request Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.24.132:8 → 192.168.24.128:0
12/04/2025-06:08:43.629243 [**] [1:2000001:0] ICMP Echo Request Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.24.132:8 → 192.168.24.128:0
12/04/2025-06:08:44.549945 [**] [1:2000001:0] ICMP Echo Request Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.24.132:8 → 192.168.24.128:0
12/04/2025-06:08:49.049435 [**] [1:2000001:0] ICMP Echo Request Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.24.132:8 → 192.168.24.128:0
```

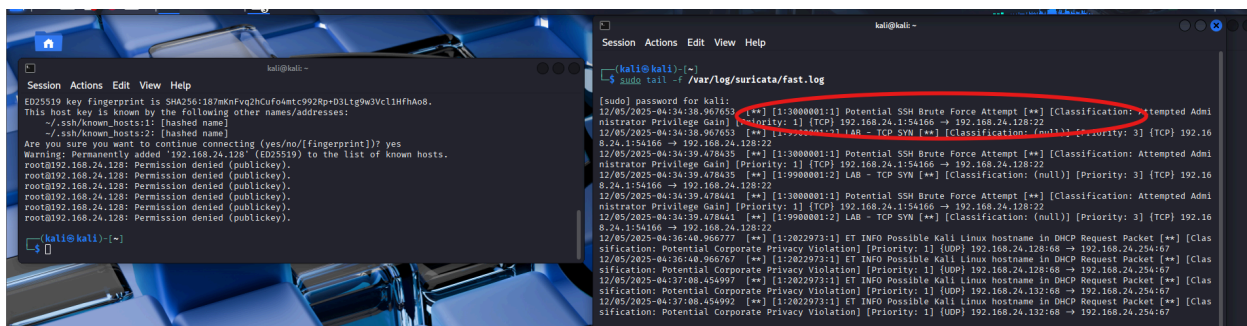
CHALLENGES :

1. Write a rule detecting SSH brute-force attacks

Rule :

alert tcp any any -> any 22 (msg:"Potential SSH Brute Force Attempt";flow:to_server,established; detection_filter:track by_src, count 5, seconds 60; classtype:attempted-admin; sid:3000001; rev:1;)

Test :



The image shows two terminal windows. The left window shows an SSH connection attempt from root@192.168.24.128 to root@192.168.24.128, which is denied. The right window shows the Suricata log output for the same event, highlighting the rule that detected the attempted administrator privilege gain.

```
kali@kali: ~
Session Actions Edit View Help

(kali@kali)~$ sudo tail -f /var/log/suricata/fast.log
[sudo] password for kali:
12/05/2025-04:34:38.967653 [**] [1:3000001:1] Potential SSH Brute Force Attempt [**] [Classification: Attempted Administrator Privilege Gain] [Priority: 1] {TCP} 192.168.24.1:54166 → 192.168.24.128:22
12/05/2025-04:34:38.967653 [**] [1:9900001:2] LAB - TCP SYN [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.24.1:54166 → 192.168.24.128:22
12/05/2025-04:34:39.478435 [**] [1:3000001:1] Potential SSH Brute Force Attempt [**] [Classification: Attempted Administrator Privilege Gain] [Priority: 1] {TCP} 192.168.24.1:54166 → 192.168.24.128:22
12/05/2025-04:34:39.478435 [**] [1:9900001:2] LAB - TCP SYN [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.24.1:54166 → 192.168.24.128:22
12/05/2025-04:34:39.478441 [**] [1:3000001:1] Potential SSH Brute Force Attempt [**] [Classification: Attempted Administrator Privilege Gain] [Priority: 1] {TCP} 192.168.24.1:54166 → 192.168.24.128:22
12/05/2025-04:34:39.478441 [**] [1:9900001:2] LAB - TCP SYN [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.24.1:54166 → 192.168.24.128:22
12/05/2025-04:36:40.966777 [**] [1:2022973:1] ET INFO Possible Kali Linux Hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 192.168.24.128:68 → 192.168.24.254:67
12/05/2025-04:36:40.966777 [**] [1:2022973:1] ET INFO Possible Kali Linux Hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 192.168.24.128:68 → 192.168.24.254:67
12/05/2025-04:37:08.454997 [**] [1:2022973:1] ET INFO Possible Kali Linux Hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 192.168.24.128:68 → 192.168.24.254:67
12/05/2025-04:37:08.454997 [**] [1:2022973:1] ET INFO Possible Kali Linux Hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 192.168.24.128:68 → 192.168.24.254:67
```

2. Detect Suspicious User-Agent

Rule :

alert http any any -> any any (msg:"Suspicious User-Agent Detected";
content:"User-Agent|3a| sqlmap"; http_header; nocase; sid:3000002; rev:1;)

Test :

The screenshot shows two terminal windows. The left window displays the output of a curl command to a web server, showing an HTML directory listing. The right window shows the Suricata logs, with a red circle highlighting the alert: [Classification: (null)] [Priority: 3] [TCP] 127.0.0.1:37190 -> 127.0.0.1:80 Suspicious User-Agent Detected [**] [Classification: (null)] [Priority: 3] [TCP] 127.0.0.1:37190 -> 127.0.0.1:80.

3. Detect Executable File Download Over HTTP

Rule :

alert http any any -> any any (msg:"Detect Executable File Download Over HTTP";
content:".exe"; http_uri; nocase; sid:3000003; rev:1;)

Test :

The screenshot shows two terminal windows. The left window displays the output of a curl command to a web server, showing a 200 status code. The right window shows the Suricata logs, with a red circle highlighting the alert: [Classification: (null)] [Priority: 3] [TCP] 127.0.0.1:37190 -> 127.0.0.1:80 Detect Executable File Download Over HTTP [**] [Classification: (null)] [Priority: 3] [TCP] 127.0.0.1:37190 -> 127.0.0.1:80.

4. Detect SQL Injection Pattern

Rule :

alert http \$EXTERNAL_NET any -> \$HOME_NET any (msg: "SQL Injection Attempt - SELECT statement"; flow: to_server, establish>

alert http any any -> \$HOME_NET any (msg: "SQL Injection Attempt - UNION statement"; flow: to_server, established; content: >

alert http any any -> \$HOME_NET any (msg: "SQL Injection Attempt - Generic SQL syntax"; flow: to_server, established; content>

Test :

The left terminal shows a curl command: `curl "http://192.168.24.128/?id=1%20select%20* HTTP/1.1" 200 -`. The right terminal shows the corresponding network traffic logs in Wireshark, highlighting the SQL injection attempt.

5. Detect Port Scanning (SYN Scan)

Rule :

alert tcp any any -> \$HOME_NET any (msg: "LAB - Port Scan (SYN threshold)"; flags: S; flow: stateless ; detection_filter: track by_src, count 20, seconds 10; sid: 9900002; rev: 1;)

Test :

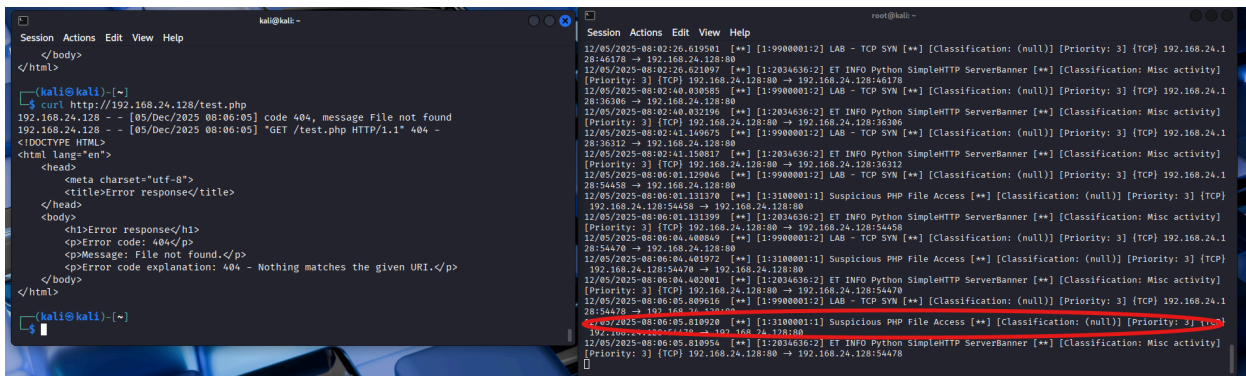
The left terminal shows an nmap scan: `sudo nmap -sS -p- 192.168.24.128`. The right terminal shows the corresponding network traffic logs in Wireshark, highlighting the port scan attempt.

6. Detect Suspicious .php Calls

Rule :

```
alert http any any -> any any (msg:"Suspicious PHP File Access"; http.method;
content:"GET"; http.uri; content:".php"; nocase; sid:3100001; rev:1;)
```

Test :

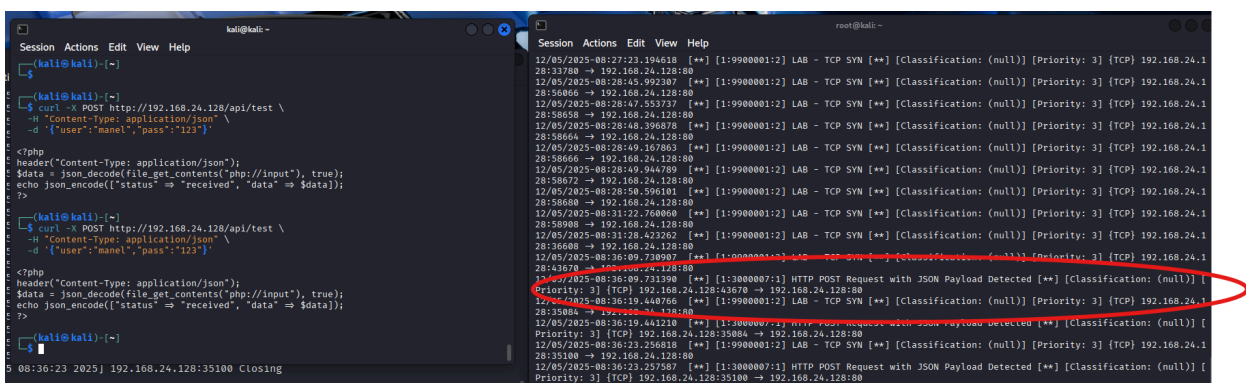


7. Detect HTTP POST Requests With JSON Payload

Rule :

```
alert http any any -> any any (msg:"HTTP POST Request with JSON Payload Detected";
flow:to_server,established; http.method; content:"POST"; http.content_type;
content:"application/json"; nocase; sid:3000007; rev:1;)
```

Test :

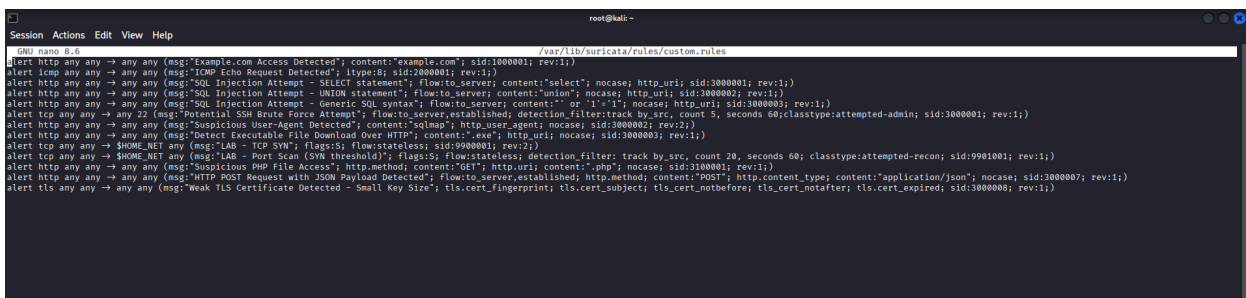


8. Detect TLS (HTTPS) Certificates With Small Key Size

Rule :

alert tls any any -> any any (msg:"Weak TLS Certificate Detected - Small Key Size";
tls.cert_fingerprint; tls.cert_subject; tls.cert_notbefore; tls.cert_notafter; tls.cert_expired;
sid:3000008; rev:1;)

My custom rules file :



```
GNU nano 8.6 /var/lib/suricata/rules/custom.rules
alert icmp any any -> any any (msg:"Example.com Access Detected"; content:"example.com"; sid:1000001; rev:1;)
alert icmp any any -> any any (msg:"ICMP Echo Request Detected"; itype:8; sid:2000001; rev:1;)
alert http any any -> any any (msg:"SQL Injection Attempt - SELECT statement"; flow:to_server; content:"select"; nocase; http.uri; sid:3000001; rev:1;)
alert http any any -> any any (msg:"SQL Injection Attempt - UNION statement"; flow:to_server; content:"union"; nocase; http.uri; sid:3000002; rev:1;)
alert http any any -> any any (msg:"SQL Injection Attempt - Generic SQL syntax"; flow:to_server; content:"'" or '1'='1"; nocase; http.uri; sid:3000003; rev:1;)
alert tcp any any -> any 22 (msg:"Potential SSH Brute Force Attempt"; flow:to_server,established; detection_filter:track by_src, count 5, seconds 60; classtype:attempted-admin; sid:3000001; rev:1;)
alert http any any -> any any (msg:"Suspicious User-Agent detected"; content:"sqlmap"; http.user_agent; nocase; sid:3000002; rev:2;)
alert http any any -> any any (msg:"Detect Executable File Download Over HTTP"; content:".exe"; http.uri; nocase; sid:3000003; rev:1;)
alert tcp any any -> SHOW_NET any (msg:"LAB - TCP SYN"; flags:S; flow:stateless; sid:9900001; rev:2;)
alert tcp any any -> SHOW_NET any (msg:"LAB - Port Scan (CVN threshold)"; flags:S; flow:stateless; detection_filter: track by_src, count 20, seconds 60; classtype:attempted-recon; sid:9901001; rev:1;)
alert http any any -> any any (msg:"Suspicious PHP File Access"; http.method; content:"GET"; http.uri; content:".php"; nocase; sid:3100001; rev:1;)
alert http any any -> any any (msg:"HTTP POST Request with JSOW Payload Detected"; flow:to_server,established; http.method; content:"POST"; http.content_type; content:"application/json"; nocase; sid:3000007; rev:1;)
alert tls any any -> any any (msg:"Weak TLS Certificate Detected - Small Key Size"; tls.cert_fingerprint; tls.cert_subject; tls.cert_notbefore; tls.cert_notafter; tls.cert_expired; sid:3000008; rev:1;)
```